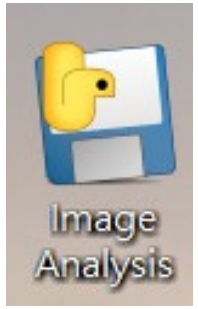
Instruction of Image Analysis GUI

GUI User Guide

Yen-Chi Chung (Angel)

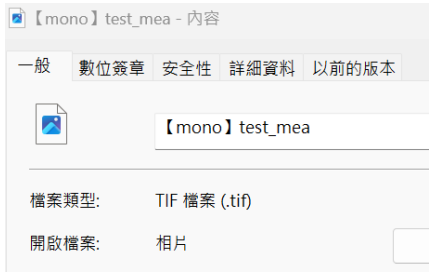
Copyright © 2026 Yen-Chi Chung (Angel)

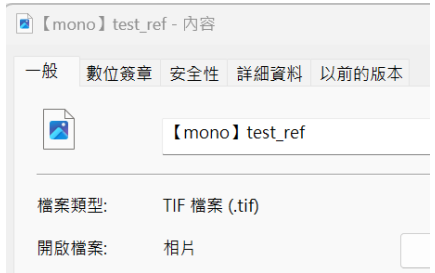
Image Analysis – GUI

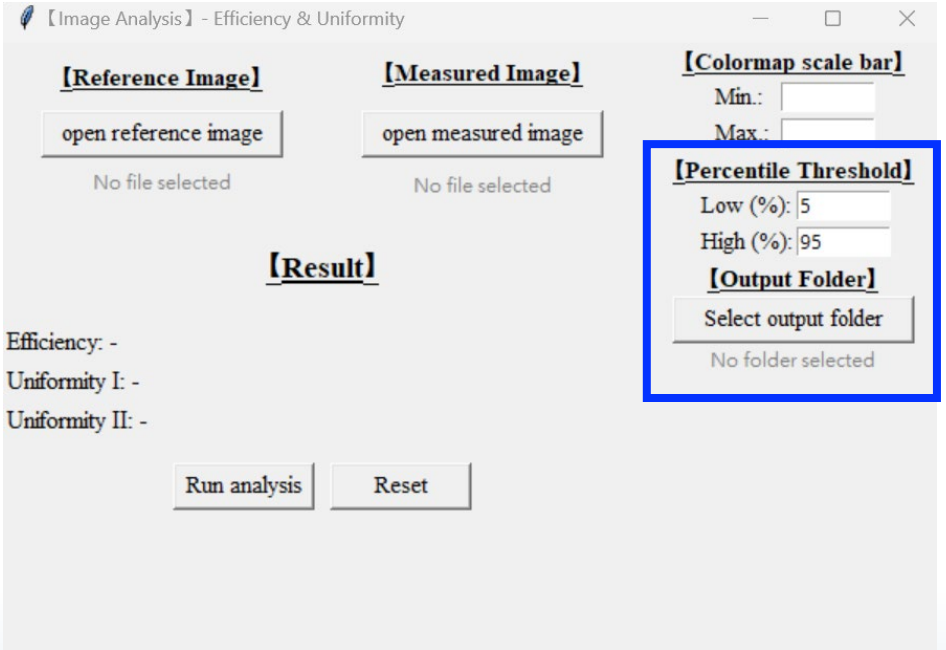


➢Image Analysis – Efficiency &Uniformity

- Method – pixel-wise division
  - Reference image : Projector → Camera
  - Measured image : Projector → Waveguide → Camera
- Condition
  - The reference image and measured image need to be gray-scale image (monochromatic)
    - ✓ Ex: .tif / .png





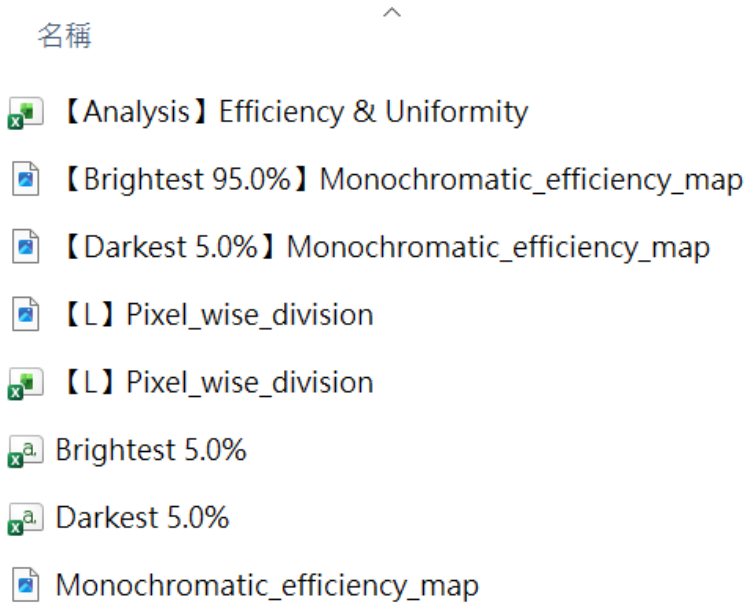


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Image Analysis – GUI

➢Image Analysis – Efficiency &Uniformity

- Output
  - Raw data
    - ✓ File name :
      - ✓ 【L】Pixel_wise_division.xlsx
  - Mapping image
    - ✓ File name :
      - ✓ Monochromatic_efficiency_map.jpg
      - ✓ 【L】Pixel_wise_division.tif
      - ✓ 【Brightest 5.0%】Monochromatic_efficiency_map.jpg
      - ✓ 【Darkest 5.0%】Monochromatic_efficiency_map.jpg
  - Statistical analysis
    - ✓ File name :
      - ✓ 【Analysis】Efficiency & Uniformity.xlsx
      - ✓ Brightest 5.0%.csv
      - ✓ Darkest 5.0%.csv

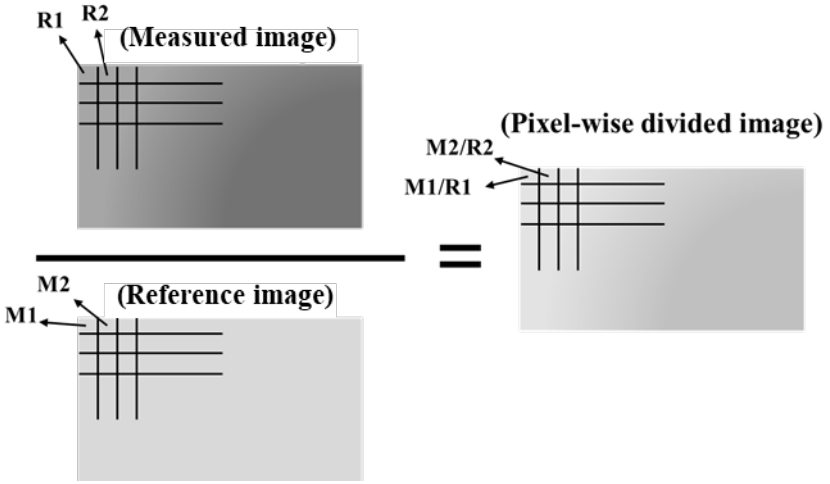


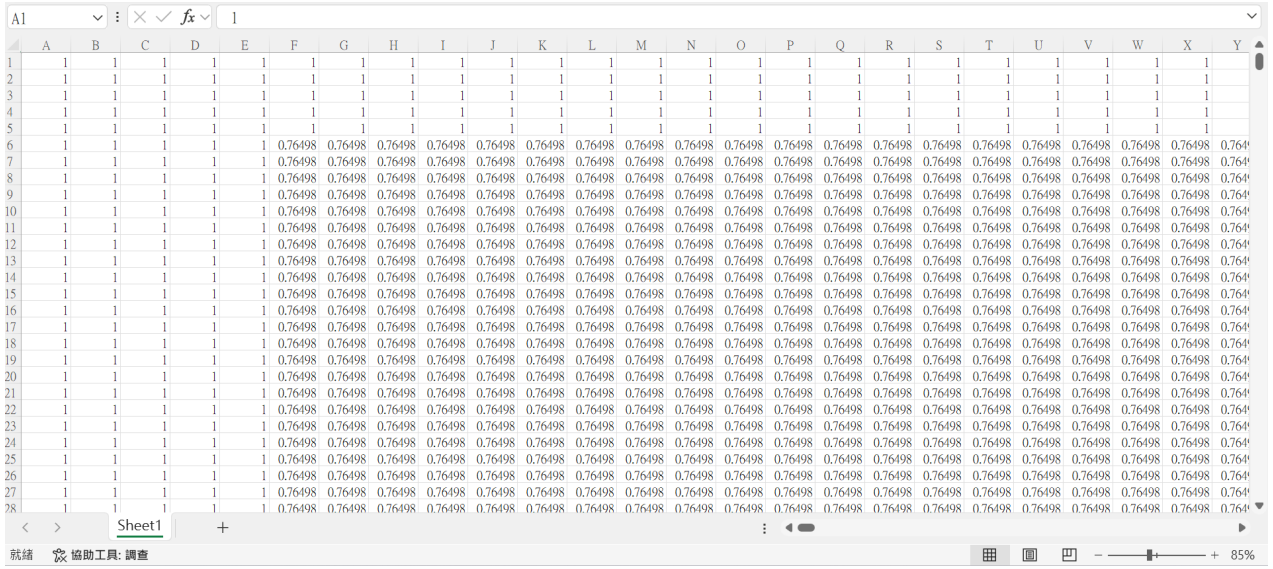
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Image Analysis – GUI

- Image Analysis – Efficiency &Uniformity
  - Raw data – 【L】Pixel_wise_division.xlsx
    - Pixel-wise division of measured image to reference image
    - It is for further detailed investigation and for further applications



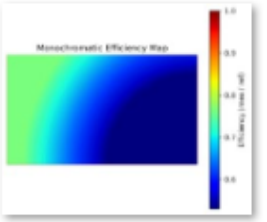




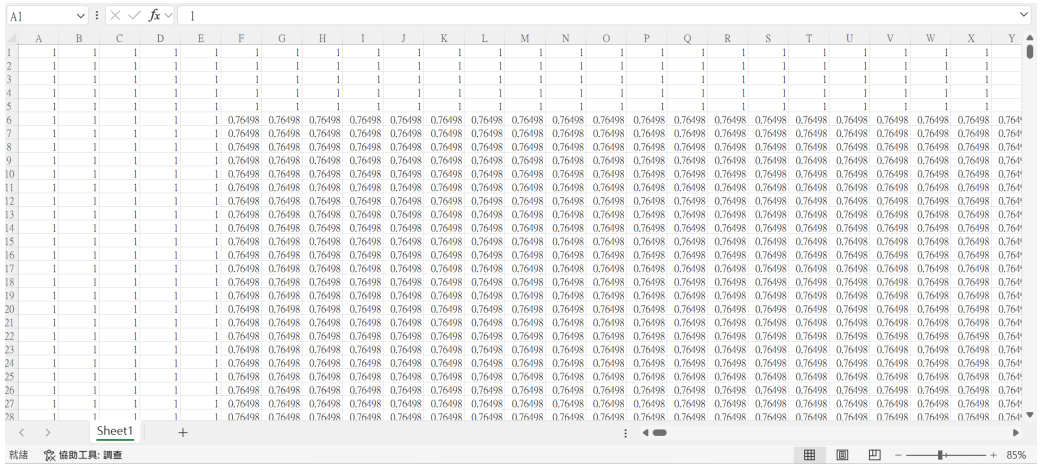
Copyright © 2026 Yen-Chi Chung (Angel)

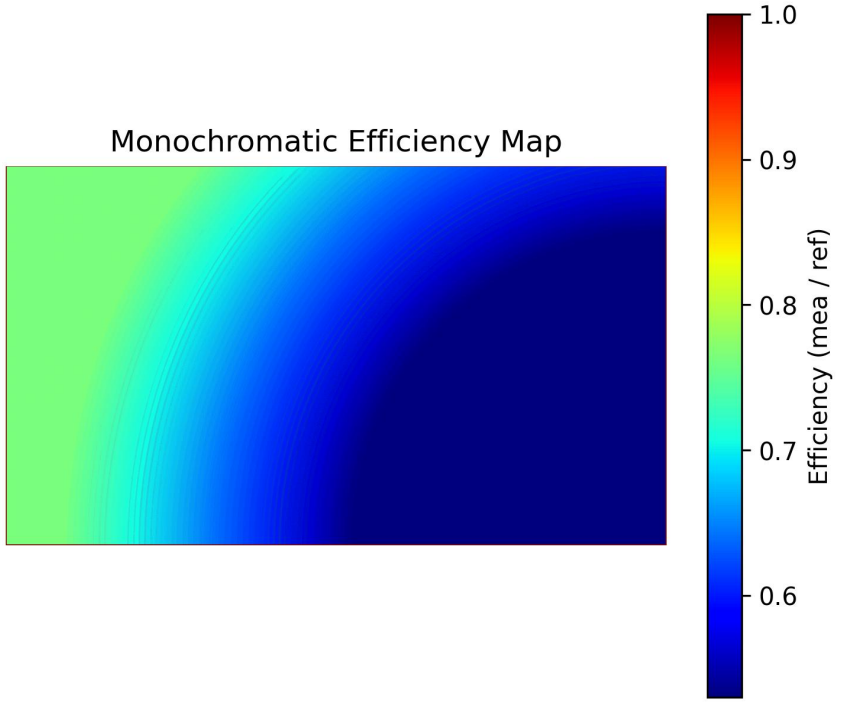
Image Analysis – GUI

- Image Analysis – Efficiency &Uniformity
  - Mapping image – Monochromatic_efficiency_map.jpg
    - Visualizing the whole efficiency based on pixel-wise division
    - It is easier to understand the whole trend and the abnormal points



Monochromatic
_efficiency_map

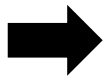




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Image Analysis – GUI

- Image Analysis – Efficiency &Uniformity
  - Mapping image – 【L】Pixel_wise_division.tif
    - Visualizing the whole efficiency based on pixel-wise division
    - It is easier to understand the whole trend and the abnormal points

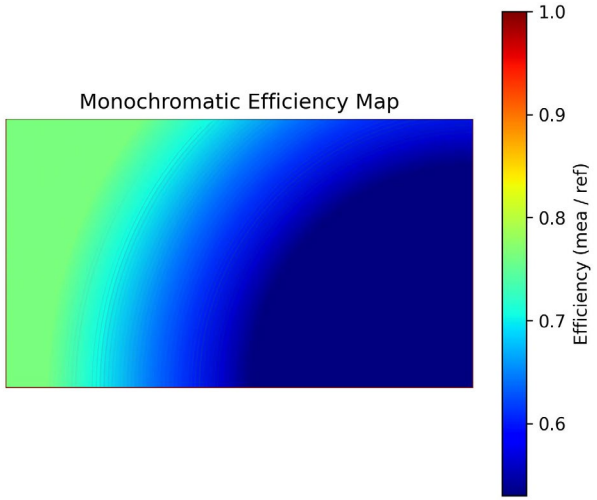


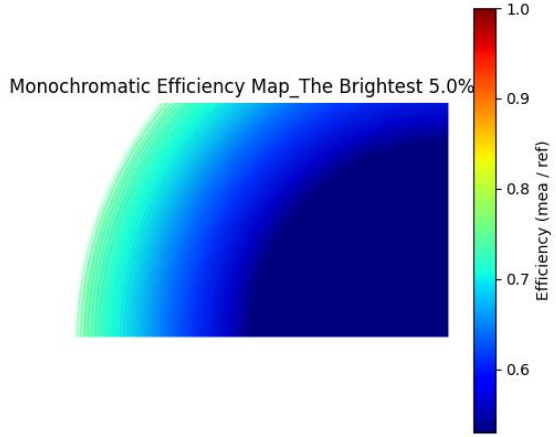
Copyright © 2026 Yen-Chi Chung (Angel)

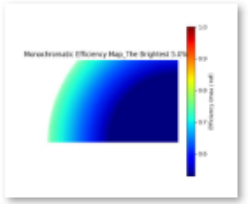
Image Analysis – GUI

➢Image Analysis – Efficiency &Uniformity

- Mapping image – 【Brightest 5.0%】Monochromatic_efficiency_map.jpg
  - Visualizing and marking the brightest 5% on the efficiency map
  - The pixel values that are over the threshold are represented by the white area
  - The mask threshold is manually adjustable by percentile value (ex: brightest 5% → 95)







【Brightest 5.0%】Monochromatic_efficiency_map

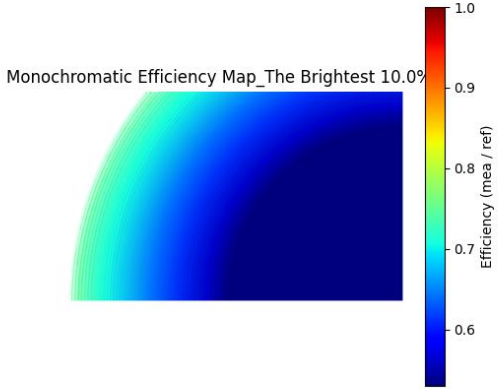
Copyright © 2026 Yen-Chi Chung (Angel)

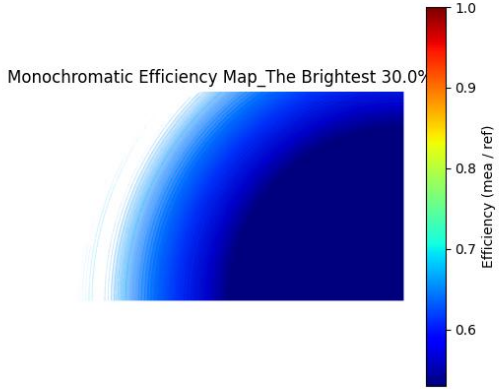
Image Analysis – GUI

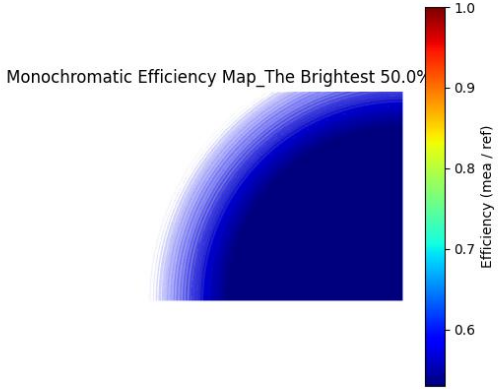
➢Image Analysis – Efficiency &Uniformity

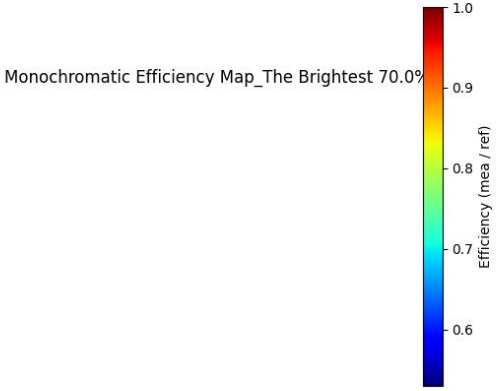
- Mapping image – 【Brightest XX%】Monochromatic_efficiency_map.jpg
  - The threshold is manually adjustable by percentile value (ex: brightest 30% → 70)
  - The masked area is represented by the white area

Ex: [0, 0.02, 0.02, 0.02, 0.02, 0.05, 0.06, 0.08, 0.08, 0.09, …….] → sorted 200 values
30% index = (200 -1) x 70% = 199 x 30% = 59.7 → the threshold is between 59th and 60th







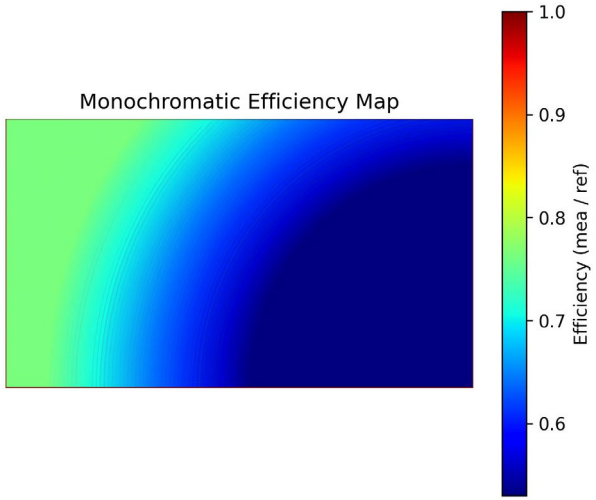


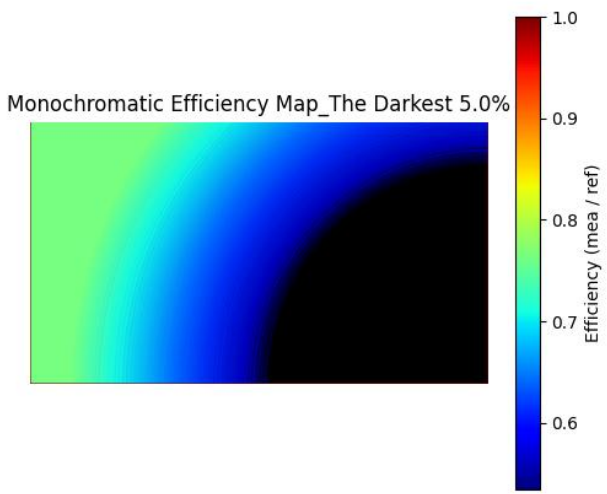
Copyright © 2026 Yen-Chi Chung (Angel)

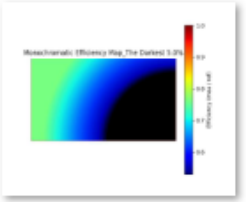
Image Analysis – GUI

➢Image Analysis – Efficiency &Uniformity

- Mapping image – 【Darkest 5.0%】Monochromatic_efficiency_map.jpg
  - Visualizing and marking the darkest 5% on the efficiency map
  - The pixel values that are below the threshold are represented by the black area
  - The mask threshold is manually adjustable by percentile value (ex: darkest 5% → 5)







【Darkest 5.0%】Monochromatic_efficiency_map

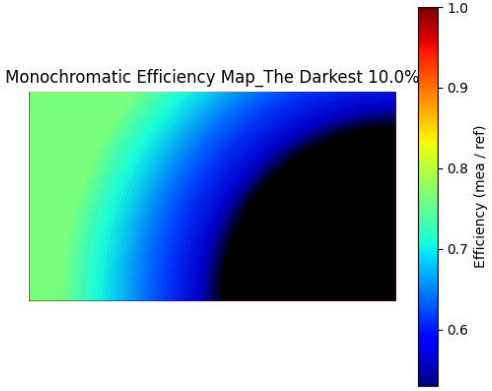
Copyright © 2026 Yen-Chi Chung (Angel)

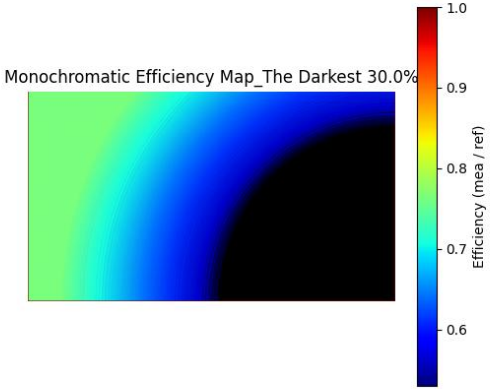
Image Analysis – GUI

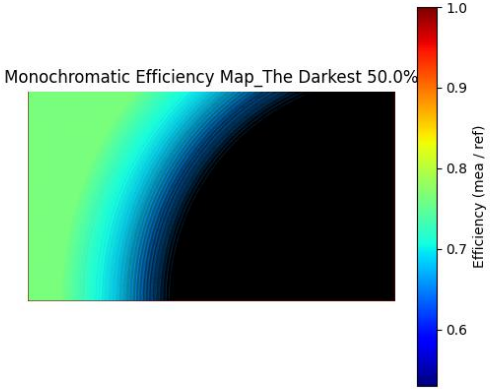
➢Image Analysis – Efficiency &Uniformity

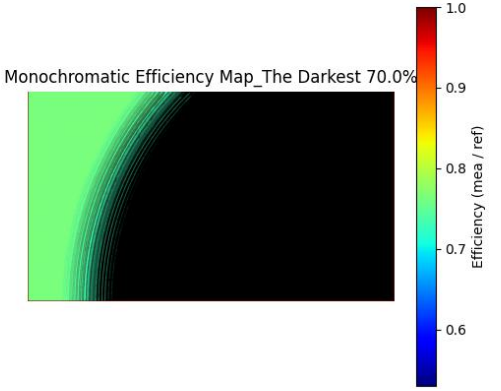
- Mapping image – 【Darkest XX%】Monochromatic_efficiency_map.jpg
  - The threshold is manually adjustable by percentile value (ex: darkest 10% → 10)
  - The masked area is represented by the black area

Ex: [0, 0.02, 0.02, 0.02, 0.02, 0.05, 0.06, 0.08, 0.08, 0.09, …….] → sorted 200 values
10% index = (200 -1) x 10% = 199 x 10% = 19.9 → the threshold is between 19th and 20th









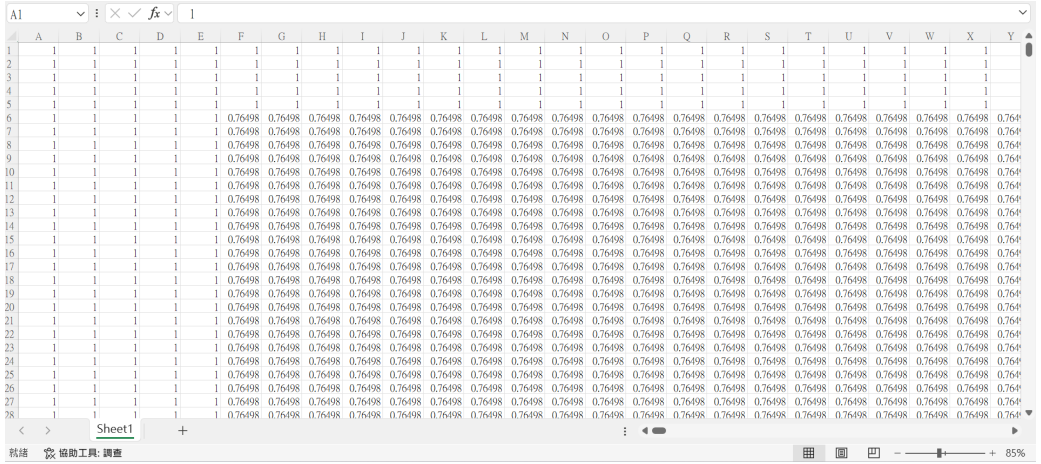
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Image Analysis – GUI

➢Image Analysis – Efficiency &Uniformity

- Statistical analysis – 【Analysis】Efficiency & Uniformity.xlsx
  - It is calculated after pixel-wise division
  - It includes average efficiency, uniformity, max, min, std, 5% max, 5% min
- Notes
  - Mean = Efficiency = average of pixel-wise division
  - Uniformity I = std / mean
  - Uniformity II = (max - mean) / mean







M6

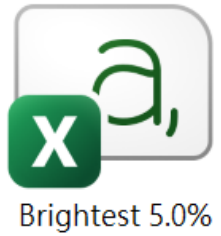
	A	B	C	D	E	F	G	H	I	J
1	Color	Mean (Efficiency)	Uniformity_I	Uniformity_II	std	Max	Min	Darkest 5%	Brightes 5%	
2	Mono	0.627	0.158	0.595	0.099	1	0.53	0.53	0.765	
3										
4										
5										

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Image Analysis – GUI

➢Image Analysis – Efficiency &Uniformity

- Statistical analysis – Brightest 5.0%.csv
  - It shows the coordinate of pixels that are over the threshold(ex: 95%)
  - The percentile can be entered manually, or the default is 95%
  - It shows the location of each marked pixels that contribute to the brightest 5%
  - The coordinate (x(col), y(row)) are the not same as plotting coordinate on Excel
    - ✓ x(colormap) = x(plot)
    - ✓ y(colormap) = -y(plot)



	A	B	C	D
1	Loc_x(col)	Loc_y(row)	ratio	
2	0	0	1	
3	1	0	1	
4	2	0	1	
5	3	0	1	
6	4	0	1	
7	5	0	1	
8	6	0	1	
9	7	0	1	
10	8	0	1	
11	9	0	1	
12	10	0	1	

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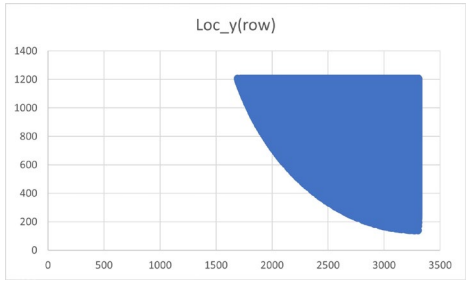
Image Analysis – GUI

➢Image Analysis – Efficiency &Uniformity

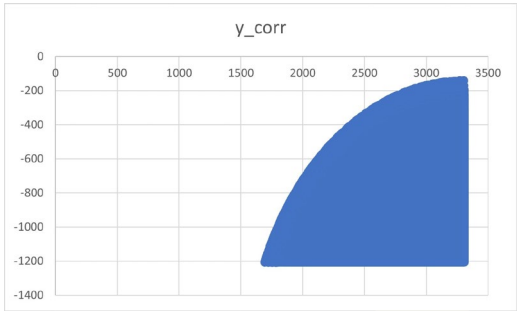
- Statistical analysis – Darkest 5.0%.csv
  - It shows the coordinate of pixels that are below the threshold(ex: 5%)
  - The percentile can be entered manually, or the default is 5%
  - It shows the location of each marked pixels that contribute to the Darkest 5%
  - The coordinate (x(col), y(row)) are the not same as plotting coordinate on Excel
    - ✓ x(colormap) = x(plot)
    - ✓ y(colormap) = -y(plot)



	A	B	C	D
1	Loc_x(col)	Loc_y(row)	ratio	
2	3268	138	0.529954	
3	3272	138	0.529954	
4	3304	138	0.529954	
5	3271	139	0.529954	
6	3274	139	0.529954	
7	3301	139	0.529954	
8	3223	140	0.529954	
9	3227	140	0.529954	
10	3233	140	0.529954	
11	3237	140	0.529954	
12	3254	140	0.529954	
13	3265	140	0.529954	



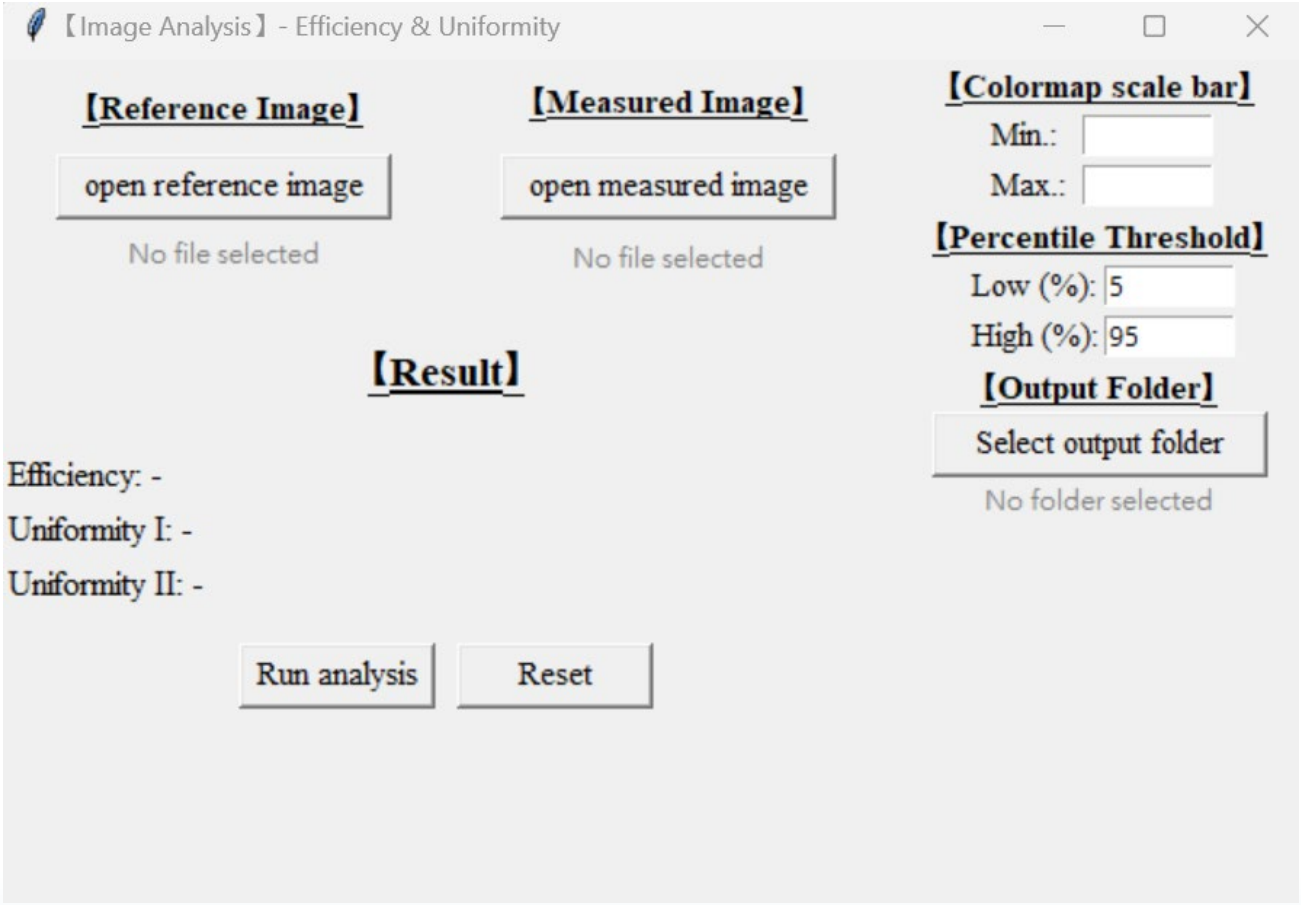
	A	B	C
1	Loc_x(col)	Loc_y(row)	y_corr
2	3268	138	-138
3	3272	138	-138
4	3304	138	-138
5	3271	139	-139
6	3274	139	-139
7	3301	139	-139
8	3223	140	-140
9	3227	140	-140
10	3233	140	-140
11	3237	140	-140
12	3254	140	-140
13	3265	140	-140



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Image Analysis – GUI

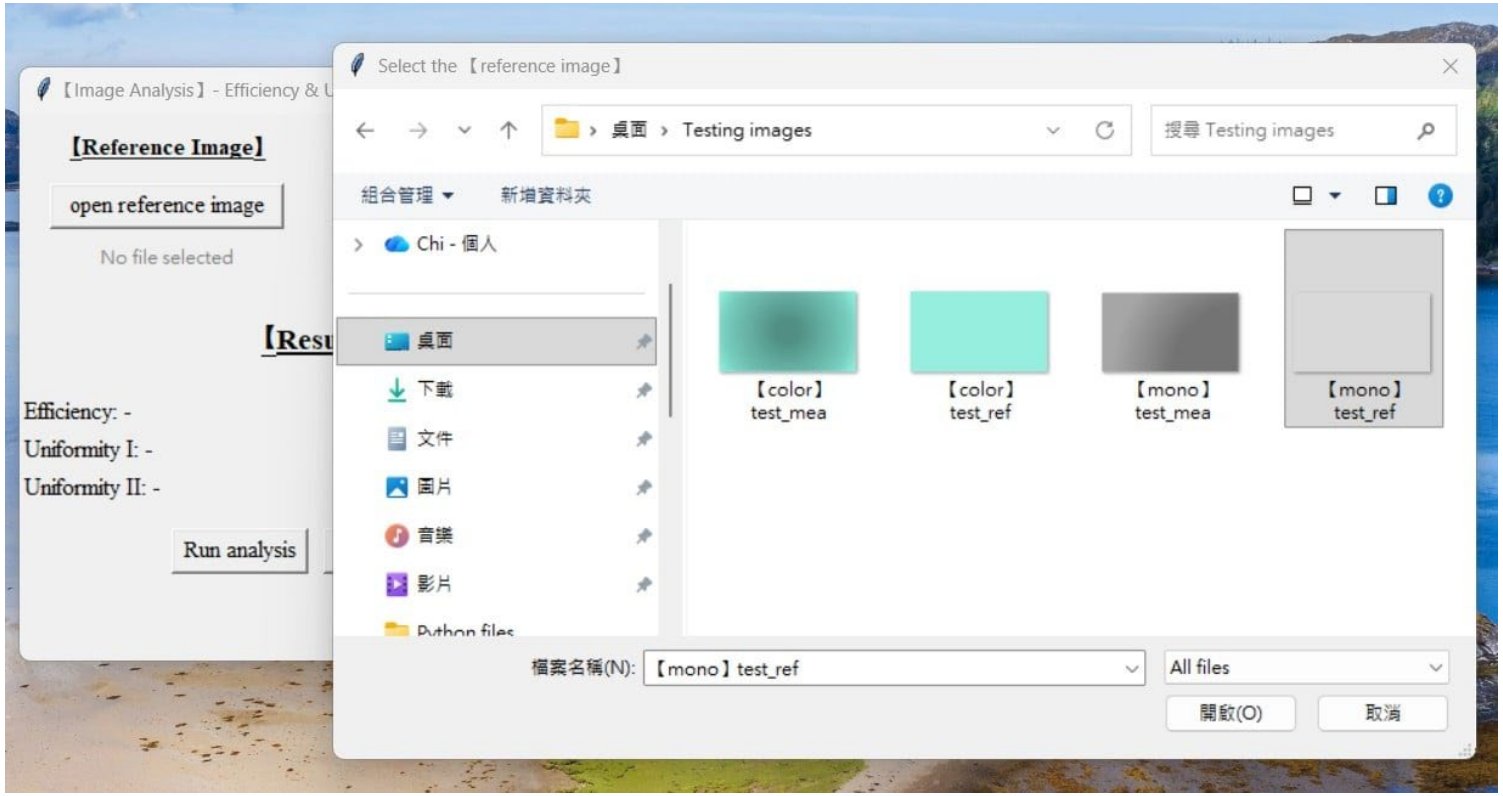
- Image Analysis – Efficiency &Uniformity
  - Graphic user-interface (GUI) for team members

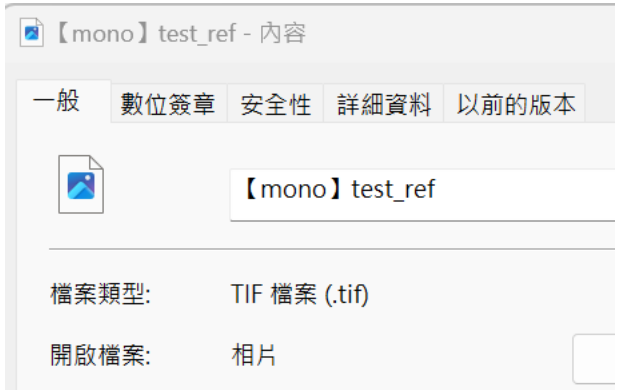


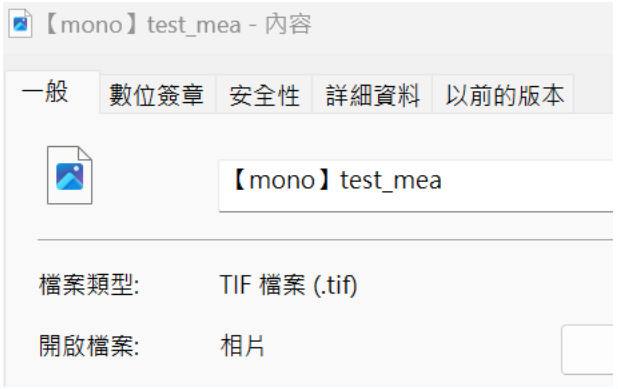
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Image Analysis – GUI

- Image Analysis – Efficiency &Uniformity
  - Select your image(.tif) for analysis
    - Reference image
    - Measured image



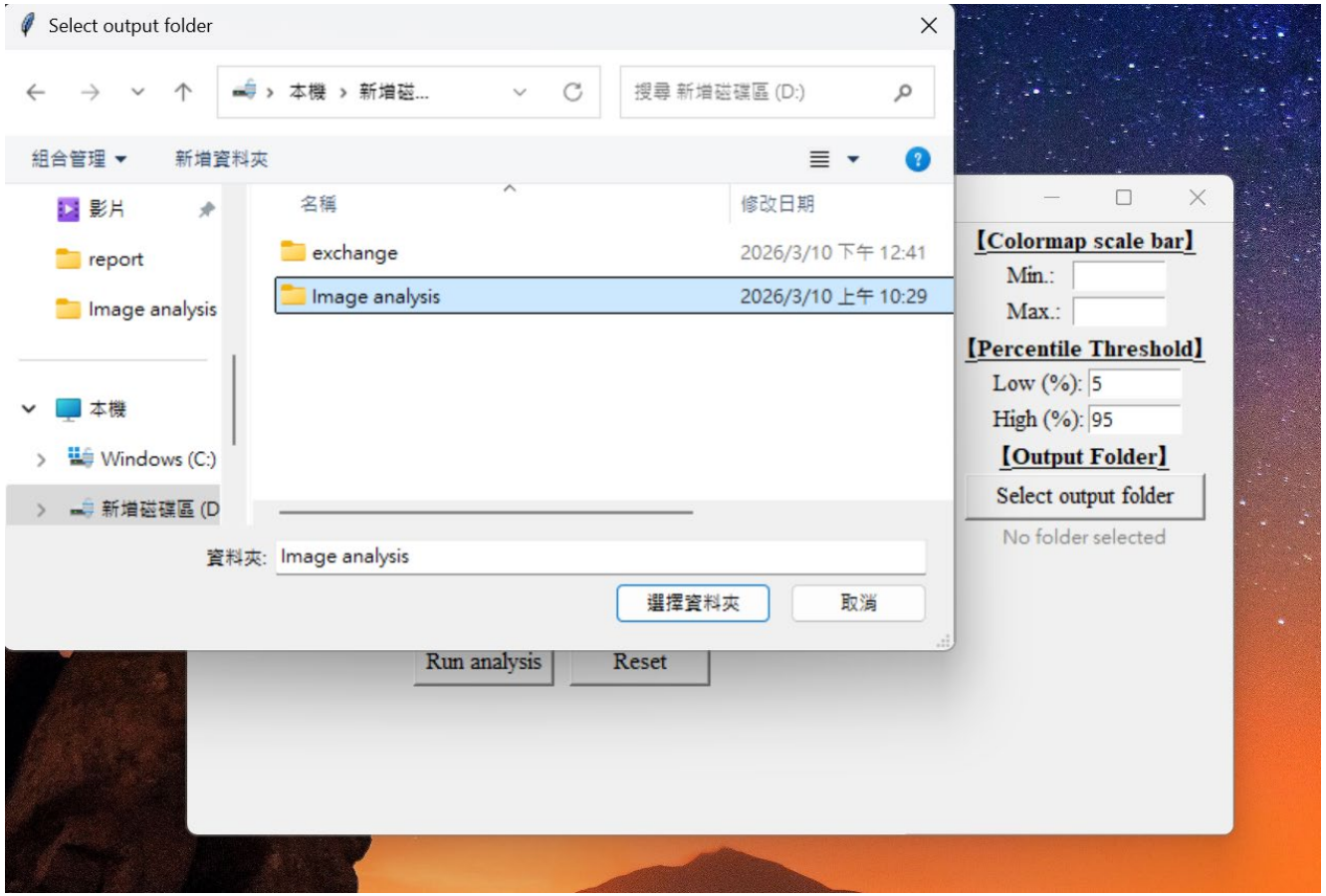




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Image Analysis – GUI

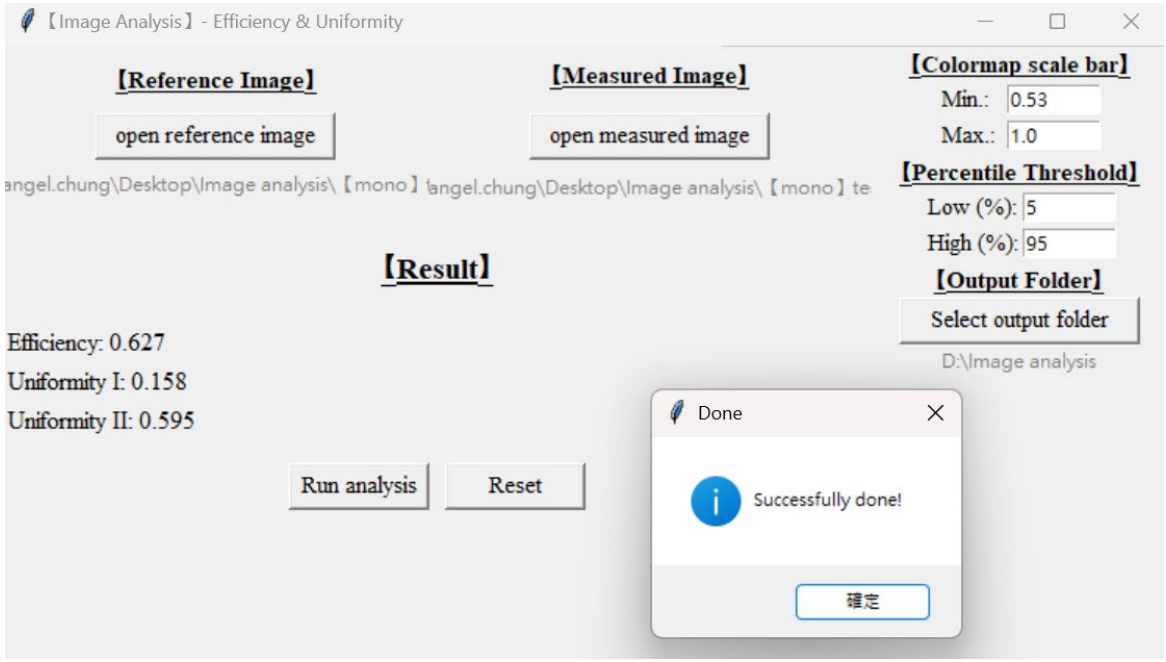
- Image Analysis – Efficiency &Uniformity
  - Select output folder



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Image Analysis – GUI

- Image Analysis – Efficiency &Uniformity
  - Press “Run analysis” and it takes about 43 sec for analysis
  - You can define the range of scale bar before ”Run analysis”
    - Colormap scale bar : default 0 ~ 1
    - Percentile threshold : default 5, 95

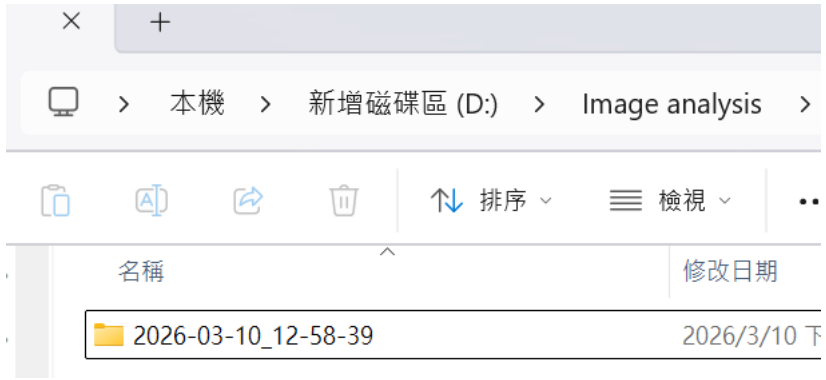


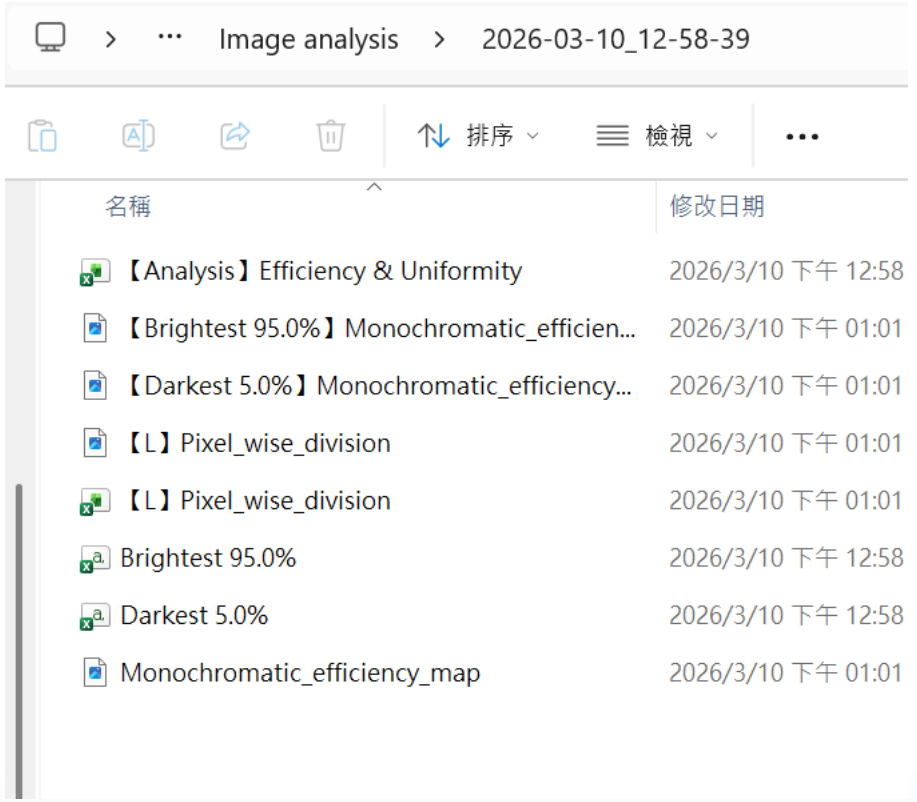
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Image Analysis – GUI

➢Image Analysis – Efficiency &Uniformity

- It creates the file named by the date under D://
- The analysis results are saved automatically in the file





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Thank you for your time.

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